Traditional RAG vs WikiLLM-Based RAG

This document compares a conventional Retrieval-Augmented Generation (RAG) system with a WikiLLM-based knowledge management architecture. While both use Large Language Models for question answering, their approaches to storing and retrieving knowledge differ significantly.

Aspect	Traditional RAG	WikiLLM-Based RAG
Knowledge Storage	Vector database with embeddings	Structured wiki pages, knowledge graph, metadata
Data Processing	Chunk documents and generate embeddings	Extract entities, facts, concepts and relationships
Primary Retrieval	Semantic similarity search	Wiki page retrieval + linked knowledge traversal
Knowledge Organization	Unstructured chunks	Structured interconnected knowledge
Context Source	Retrieved text chunks	Curated wiki pages with relationships
Reasoning	Limited to retrieved chunks	Reasoning across linked concepts
Updates	Re-embed changed documents	Incrementally update wiki knowledge
Explainability	Moderate	High (knowledge pages and links are visible)
Best For	Document search and Q&A	Knowledge management, enterprise AI assistants, do

Architecture Comparison

Traditional RAG

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User → Upload PDF → Chunking → Embedding Model → Vector Database →
Retriever → LLM → Answer
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WikiLLM-Based RAG

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User → Upload PDF → Document Parser → LLM Knowledge Extraction →
Wiki Page Generator → Knowledge Repository →
Knowledge Linking → Context Builder → LLM → Answer
```

Advantages of WikiLLM

- Converts documents into reusable structured knowledge.
- Creates relationships between entities and concepts.
- Reduces duplicate context and improves consistency.
- Supports knowledge reuse across multiple documents.
- Easier to maintain and explain than raw vector chunks.
- Can combine with vector search for hybrid retrieval.

When to Choose Each Approach

Traditional RAG: Best for quick document search, FAQs, and retrieval over static documents.

WikiLLM-Based RAG: Best for enterprise knowledge bases, documentation portals, research assistants, technical manuals, and long-term knowledge management systems.